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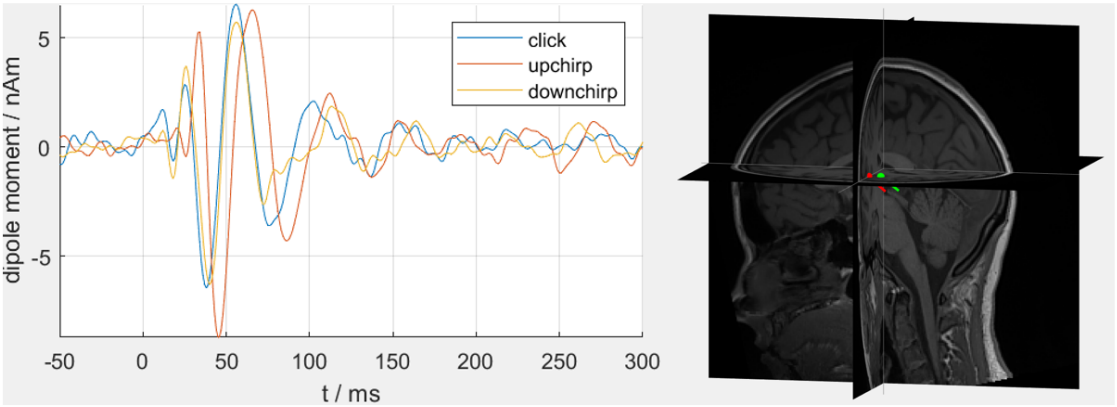
FORTGESCHRITTENENPRAKTIKUM PHYSIK (FPR-B) / TEIL BLOCKPRAKTIKUM PSYCHOPHYSIK, NEUROSENSORIK UND AUDITORISCHE SIGNALVERARBEITUNG

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DEPARTMENT OF MEDICAL PHYSICS AND ACOUSTICS

The representation of peripheral neural activity in the middle-latency evoked field of primary auditory cortex in humans

Magnetoencephalography (MEG) project - version 1.0



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Acronyms

A1 Auditory Cortex. 6
ABRs Auditory Brainstem Responses. 3
AEFs Auditory Evoked Fields. 6
AEPs Auditory Evoked Potentials. 6

BIDS Brain Imaging Data Structure. 15
BM Basilar Membrane. 3

ECDs Equivalent Current Dipoles. 5, 10, 13
ECG Electrocardiography. 11, 19
EEG Electroencephalography. 4, 5, 6, 9, 10, 11, 12
EFs Evoked Fields. 5, 6
EOG Electrooculography. 11, 19
EPs Evoked Potentials. 5, 6

HPI Head Position Indicator. 11, 19

MEG Magnetoencephalography. 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 16, 17, 18, 19
MLAEFs Middle-Latency Auditory Evoked Fields. 3, 6, 11, 13, 15, 16, 17, 19
MNI Montreal Neurological Institute. 13, 14, 17, 18
MRI Magnetic Resonance Imaging. 9, 11, 12, 13, 14, 16, 17, 18, 19
MSR magnetically shielded room. 11

PAC Primary Auditory Cortex. 3, 4, 7, 14

SQUIDs Superconducting Quantum Interference Devices. 5, 7
SSS Signal Space Separation. 11
SVD singular value decomposition. 14, 18

1 Introduction/ Motivation

This practical session will explore the neural processing of transient auditory stimuli, specifically focusing on how peripheral synchronization influences cortical evoked responses, building upon the foundational work by Rupp et al. (2002).
The auditory system processes sound through the cochlea, where mechanical properties, such as the exponential decrease in Basilar Membrane (BM) stiffness from base to apex, lead to spatial dispersion of sound components. Consequently, the response to broadband transient stimuli like clicks is not uniformly synchronous across frequency channels. Lower frequency components experience a propagation delay, resulting in delayed activation compared to higher frequencies. Therefore, typical click-evoked responses predominantly reflect the synchronized activity of high-frequency channels at the basal end of the cochlea.
Prior research has explored the use of specially designed frequency-modulated chirps to enhance neural synchrony across the cochlea. Up-chirps, characterized by an increasing instantaneous frequency, are optimized to compensate for cochlear dispersion. This compensation leads to significantly enhanced wave V amplitudes in Auditory Brainstem Responses (ABRs) and a latency shift compared to responses elicited by down-chirps or standard clicks. These findings indicate that cochlear phase delays are preserved at least up to the level of the inferior colliculus.
However, the extent to which this peripheral synchrony propagates to the Primary Auditory Cortex (PAC) and influences the morphology and timing of cortical evoked responses has been investigated by Rupp et al. (2002).

1.1 Aims and Objectives

This study aims to investigate the role of the PAC in the processing of transient sounds, with a particular focus on how peripheral neural synchrony impacts PAC activation and the morphology of middle-latency components. To address this, the present study will utilize Magnetoencephalography (MEG) to examine how varying degrees of peripheral neural synchrony, manipulated through different chirp and click stimuli, influence activation in the PAC. The experiment is designed to test two specific hypotheses regarding Middle-Latency Auditory Evoked Fields (MLAEFs) in the Primary Auditory Cortex:

1. A higher degree of peripheral synchronization, achieved through specially designed auditory stimuli, will result in a larger N19m-P30m complex in the MEG recordings.
2. The latency of this cortical complex (N19m-P30m) will be dependent on the stimulus type.

By comparing the MLAEFs responses elicited by up-chirps, down-chirps, and clicks, this investigation will assess whether the influence of peripheral phase alignment across frequency channels reflects the preservation of cochlear timing information at the cortical level, reproducing directly the findings in Rupp et al. (2002).

2 Theory

Understanding this experiment requires an interdisciplinary background. This chapter, therefore, provides a foundational overview of the key topics and theoretical concepts necessary to comprehend the study's aims. As this introduction is selective and not exhaustive, readers are encouraged to pursue further research in specific areas for a more in-depth understanding.

2.1 The Cochlea

The cochlea is a part of the inner ear responsible for analyzing the frequencies of incoming acoustic signals. This function is primarily enabled by the Basilar Membrane, a key component within the cochlea. The BM exhibits a crucial physical characteristic: it becomes progressively wider and less stiff from its base (near the entrance) toward its apex (the far end). Consequently, an incoming sound signal containing multiple frequencies causes traveling waves whose peaks occur at different locations along the BM, depending on the frequency. High frequencies achieve maximum displacement near the base, while low frequencies resonate near the apex. Crucially, due to this structure, the response to low frequencies reaches its peak later than the response to high frequencies.
An up-chirp stimulus is specifically designed to compensate for these propagation delays, which arise from the BM's stiffness gradient. The underlying physical model often used to describe this stiffness $c(x)$ as a function of position x along the cochlea is based on de Boer (1980), with parameters for the human cochlea:

$$c(x) = C_0 e^{-\alpha x}$$

Here, $C_0 = 10^4$ N cm^{-3} represents the stiffness at the base, and $\alpha = 3.0$ cm^{-1} determines the rate of stiffness change (Rupp et al., 2002). The fundamental hypothesis is that by starting with low frequencies and sweeping upwards, the up-chirp compensates for the natural travel time differences along the cochlea. By presenting lower frequencies earlier, the aim is to improve the temporal alignment of responses across different frequency regions, potentially leading to more synchronized neural activity.
This study investigates whether this peripheral synchronization effect, demonstrated at the brainstem level, is preserved further up the auditory pathway, specifically within the PAC, located in the temporal lobe.

2.2 Acoustic Signals

Acoustic signals are pressure variations in the air, or another medium, that propagate as sound waves and are detected by the auditory system. These signals are characterized by physical properties such as frequency, amplitude, phase, and duration. These physical attributes correspond to our perceptual experiences of pitch, loudness, and timbre, respectively.
In the context of auditory neuroscience, these signals act as the initial stimuli, triggering a cascade of neural processing. This journey begins with the transduction of sound in the cochlea, progresses through various stages including the auditory nerve, brainstem, and thalamus, and ultimately reaches the Primary Auditory Cortex. A crucial goal is to understand how these signals are encoded and represented, particularly during the early-to-middle cortical processing stages (typically 10–50 ms post-stimulus), as these middle-latency responses offer a window into how peripheral neural activity influences cortical dynamics. We delve deeper into these responses in chapter 2.4.
To investigate how the auditory system processes temporal features and handles inherent physiological delays, specific acoustic stimuli with distinct structures that vary in time are often employed. Figure 1 illustrates the temporal waveforms of three such stimuli commonly used in this field: a standard click, an up-chirp, and a down-chirp.
As depicted in Figure 1, these signals have unique temporal profiles, shown with normalized amplitudes to facilitate comparison. The click provides a sharp, broadband, and near-instantaneous onset. The up-chirp features an instantaneous frequency that increases exponentially over time; such stimuli are often designed to compensate for known processing delays within the cochlea, potentially enhancing neural synchrony. In contrast, the down-chirp exhibits a decreasing frequency sweep, which can exaggerate these inherent delays. By analyzing the brain's responses to these different signals, we gain valuable insights into how critical features like neural synchrony and temporal integration are managed within the auditory pathway.

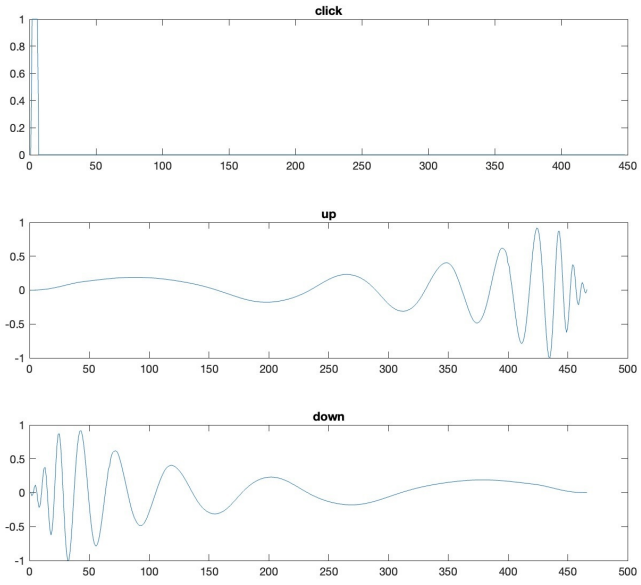


Figure 1: Normalized amplitude of click, up-chirp, and down-chirp signals over time. These distinct temporal structures allow researchers to probe how the auditory system handles onset timing and frequency sweeps.

2.3 Electrophysiology

Electrophysiology refers to the study of electrical activity generated by neurons in the nervous system. Methods like Magnetoencephalography (MEG) and Electroencephalography (EEG) allow us to observe this activity non-invasively, capturing the brain's dynamics with millisecond temporal resolution. This makes them particularly valuable for studying rapid processes such as auditory perception.
Although both MEG and EEG measure brain activity in real-time, they differ in their approach and sensitivity:

- **EEG** records electrical potentials on the scalp using electrodes. It is relatively easy to deploy and cost-effective. However, electrical signals are distorted by the varying conductivities of the skull, scalp, and cerebrospinal fluid, which can complicate the precise localization of neural sources.

- **MEG** measures the magnetic fields produced by the same underlying neuronal currents. These fields are less affected by intervening tissues, typically offering improved spatial resolution and more accurate source modeling compared to EEG.

To understand the origin of these electrophysiological signals, particularly for MEG, it is essential to consider their primary source: pyramidal neurons.

Pyramidal neurons (Figure 2) are a common type of excitatory cell found in the cerebral cortex, the outermost layer of the brain involved in sensory processing, cognition, and motor control. Named for their characteristic pyramid-like shape, they consist of a cell body (soma), a long apical dendrite extending towards the cortical surface, multiple basal dendrites spreading laterally, and an axon projecting to other brain areas.

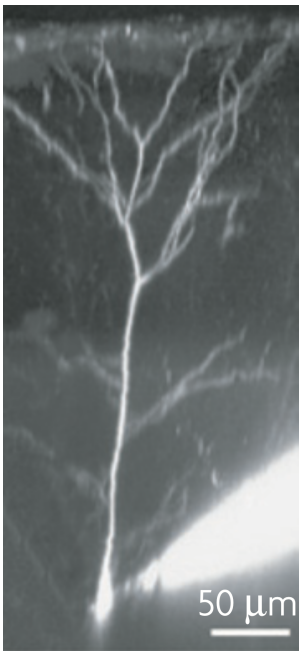


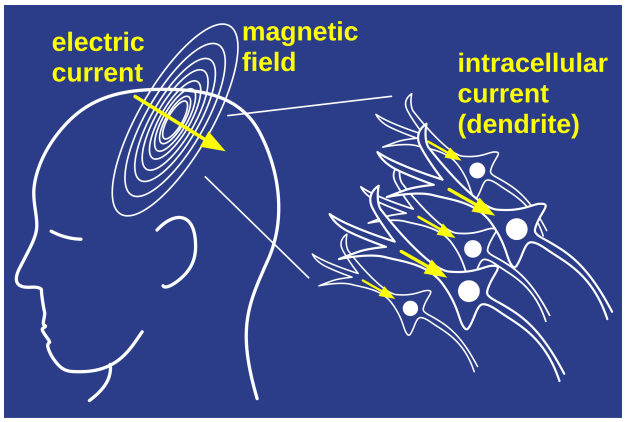
Figure 2: Pyramidal neurons, with their distinct shape and alignment, are the primary source of MEG ans EEG signals.

Within the cortex, these neurons are often organized in columns, maintaining a relatively uniform orientation with their apical dendrites generally aligned perpendicular to the cortical surface. When pyramidal neurons receive synaptic input, postsynaptic currents flow along their dendrites. This flow of ions creates an Equivalent Current Dipoles (ECDs) oriented along the dendrite's main axis.

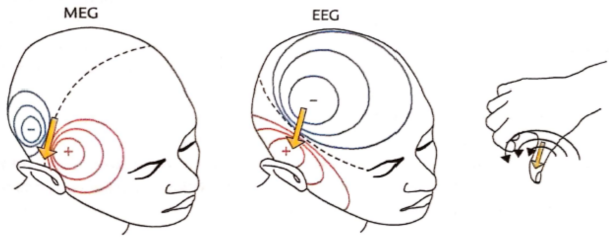
Although the field from a single neuron is far too weak to detect, when large groups of pyramidal neurons (tens of thousands) are activated synchronously, their individual current dipoles sum up. This collective activity produces a measurable macroscopic magnetic field, as illustrated in Figure 3. Although extremely weak (femtoteslas), these fields can be detected outside the head using highly sensitive sensors like Superconducting Quantum Interference Devices (SQUIDs).

Due to the physics of magnetic fields and the spherical-like nature of the head, MEG is most sensitive to currents flowing tangentially to the scalp. Since pyramidal neurons are oriented perpendicularly to the cortex, this means that the strongest MEG signals originate from neurons located in the walls of cortical folds, known as sulci. Here, their orientation results in a predominantly tangential current flow relative to the overlying scalp and sensors. In contrast, neurons at the crown of a gyrus (a cortical ridge) produce primarily radial currents, whose magnetic fields tend to cancel out and are much harder to detect externally.

Thus, the unique morphology, columnar organization, and alignment of pyramidal neurons, especially those within the cortical sulci, make them the dominant source of the magnetic fields recorded by MEG. Understanding these properties is fundamental to interpreting MEG data and accurately modeling the underlying neural activity.



(a) Synchronous activity in pyramidal neurons generates postsynaptic currents (yellow arrows) and an associated magnetic field (circular lines), detectable by MEG.



(b) Relationship between an equivalent current dipole in the auditory cortex and MEG/EEG signal distributions. Yellow indicates the dipole. MEG isofield lines and EEG isopotential lines are shown, both zero along the dashed line. MEG signs indicate magnetic flux direction (positive: outward, negative: inward); EEG signs indicate scalp potential polarity. The MEG field follows the right-hand rule (ampere's law): the thumb indicates current direction, fingers show the circular magnetic field. Figure taken from Hari and Puce (2017).

Figure 3: Macroscopic signal generation for MEG and EEG.

2.4 Evoked potentials and fields

Evoked Potentials (EPs) and Evoked Fields (EFs) are neurophysiological responses that are time-locked to specific external stimuli, serving as fundamental tools for investigating sensory processing in the human brain.

While EPs are measured using EEG as voltage fluctuations on the scalp, EFs are recorded using MEG as the magnetic counterparts. Both reflect the summation of synchronous postsynaptic activity in cortical pyramidal neurons, providing precise temporal resolution of neural responses (see Section 2.3).
In auditory neuroscience, Auditory Evoked Potentials (AEPs) and Auditory Evoked Fields (AEFs) are extensively used to examine the neural encoding and cortical representation of sound. Figure 4 illustrates the major AEPs. These responses are typically categorized by latency:

- **Short-latency responses (<10 ms)**: Originating from the auditory nerve and brainstem.
- **Middle-latency responses (10–50 ms)**: Believed to reflect thalamocortical input and early cortical processing.
- **Long-latency responses (>50 ms)**: Associated with higher-order cognitive functions.

Middle-latency components, such as the Pa and Nb waves, are particularly valuable for studying early cortical dynamics in the Auditory Cortex (A1). When recorded with MEG, these are specifically termed Middle-Latency Auditory Evoked Fields.

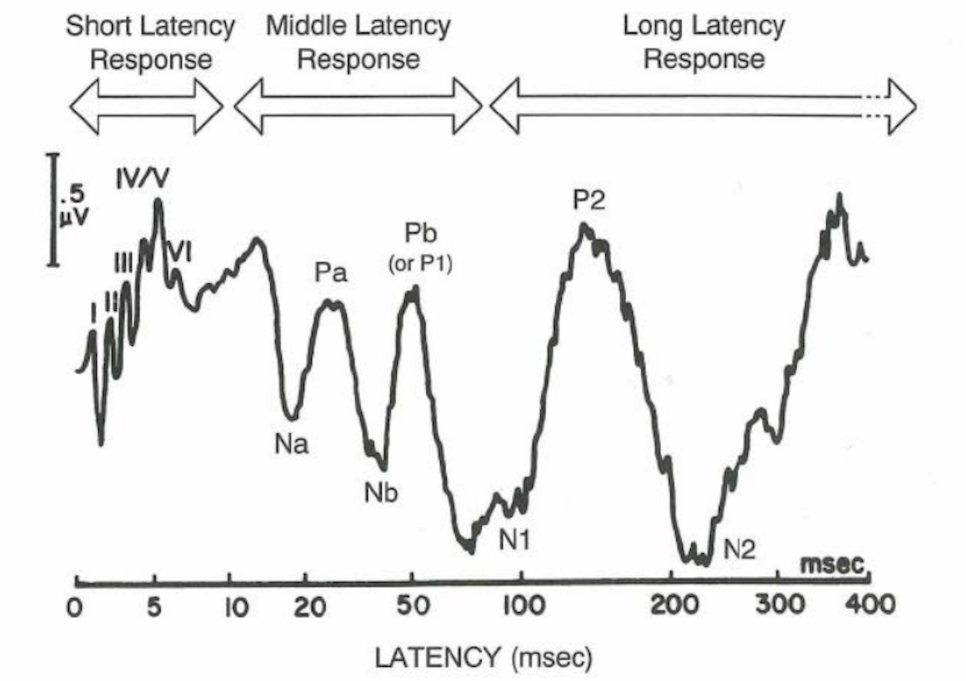


Figure 4: Major AEPs on a logarithmic scale. Adapted from Gelfand (2009).

The study of these middle-latency evoked responses using MEG allows for non-invasive investigation of how sound-related activity is represented and transformed within cortical areas, offering high temporal and spatial resolution. This line of research provides critical insights into the hierarchical organization of auditory processing, linking peripheral neural encoding with central representations in the auditory cortex. Understanding this relationship is fundamental for both basic neuroscience and clinical applications, such as assessing auditory pathway integrity in various neurological and hearing disorders.
MEG-based measurements capture the timing, strength, and spatial distribution of brain responses, offering a powerful tool to investigate auditory perception, neural encoding of sound features, and the functional organization of the auditory cortex.
In this study, we investigate the characteristic patterns of neural activity along the auditory pathway by presenting auditory stimuli, such as clicks and chirps, and analyzing their associated MLAEFs.

2.5 Magnetoencephalography

MEG is a non-invasive neuroimaging technique that measures the tiny magnetic fields generated by neural activity in the brain. These magnetic fields are produced by intracellular electrical currents, primarily in the apical dendrites of pyramidal neurons oriented perpendicular to the cortical surface (see Figure 2). MEG offers excellent temporal resolution (on the order of milliseconds) and good spatial resolution, making it particularly well-suited for studying fast neural dynamics like those involved in sensory processing. As mentioned in Chapter 2.3, MEG signals are less affected by the conductive properties of the skull and scalp compared to EEG,

allowing for more precise source localization. In auditory research, MEG is commonly used to investigate evoked fields in response to acoustic stimuli, providing information on the timing and spatial organization of activity in auditory cortical areas, including the PAC.

2.5.1 MEG Sensors

MEG systems use highly sensitive sensors, so-called SQUIDs, to detect the brain's extremely weak magnetic fields. To minimize environmental magnetic noise, MEG measurements are typically performed in a magnetically shielded room (MSR). Figure 5 shows the sensor helmet of the Elekta (now MEGIN) Neuromag Triux MEG system used in Oldenburg.

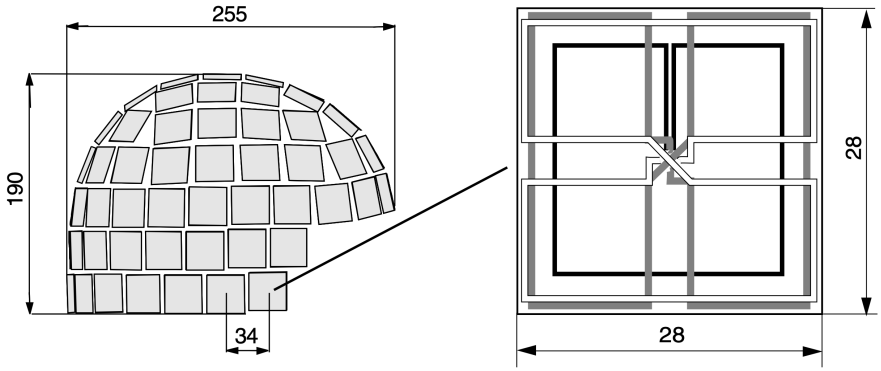


Figure 5: Elekta Neuromag Triux MEG sensor helmet. The left schematic shows the detector array in a side view. On the right a triple sensor detector unit is displayed. Dimensions are in millimeters. Each sensor unit (102 in total, 306 channels in total) consists of one magnetometer and two orthogonal planar gradiometers. Figure taken from *Elekta Neuromag® TRIUX User's Manual* (2017).

Magnetometers are MEG sensors that measure the absolute strength of the magnetic field at a single location. They are highly sensitive and can detect very weak magnetic fields from neural activity. However, because they also detect magnetic noise from external sources and the environment, they can be more susceptible to interference.
Gradiometers measure the difference in magnetic fields (gradient) between two closely spaced points. This configuration allows them to effectively suppress uniform background noise and enhance sensitivity to nearby neural sources. Gradiometers are particularly useful for detecting localized brain activity with greater precision and are often favored in environments where minimizing external noise is essential. The sensitivity profile of these gradiometers can be seen in Figure 6. For example, planar gradiometers used by the Elekta Neuromag Triux system are most sensitive directly above the source.

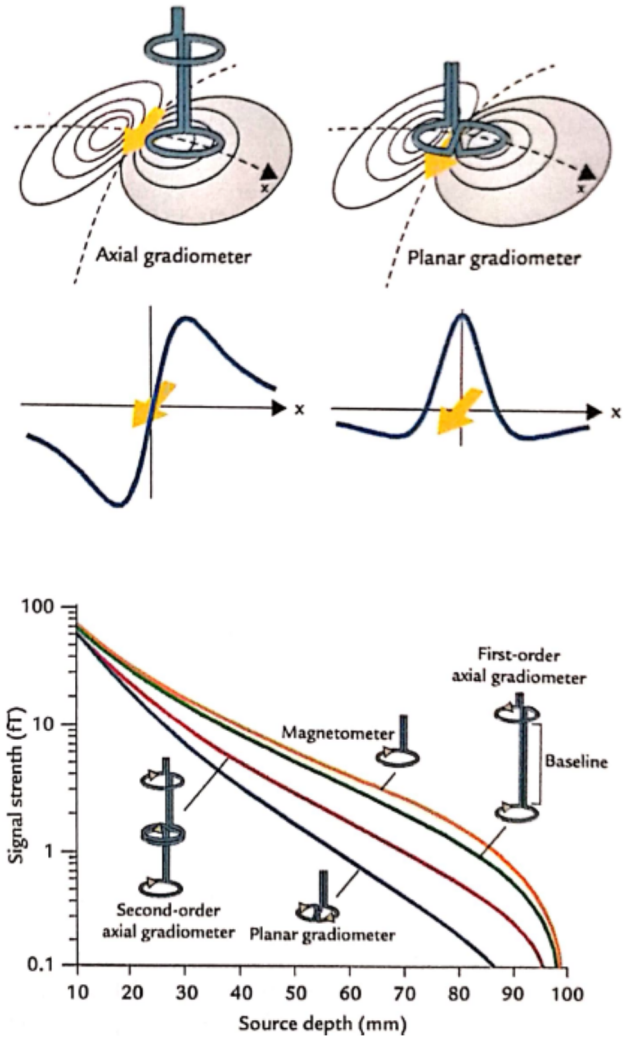


Figure 6: Influence of different MEG sensor types (flux transformers) on measured magnetic field distributions and signal strengths. The **top panel** compares axial and planar gradiometers, showing that they detect the peak signal at different locations relative to the field distribution generated by a current dipole (yellow arrow). Axial gradiometers are most sensitive above the two extrema (positive and negative peaks) of the dipolar pattern (same holds for magnetometers), whereas planar gradiometers are most sensitive directly above the source. The **middle panel** shows signal strengths plotted along a line x passing over the source, orthogonal to the current dipole's orientation; the left plot corresponds to the axial gradiometer, and the right plot to the planar gradiometer. The **bottom panel** displays signal strength (on a logarithmic scale) as a function of source depth within a 100 mm radius sphere. Magnetometers measure the strongest signals, while planar gradiometers measure the weakest signals for sources at all depths. No sensor type is sensitive to a source located precisely at the center of the sphere, as such a source produces no measurable external magnetic field. Figure taken from Hari and Puce (2017).

2.5.2 Forward Modeling, Inverse Modeling, and Lead Fields

In the context of MEG and EEG source localization, two essential computational steps are required: forward modeling and inverse modeling.
Forward modeling is the process of calculating the signals expected at the MEG/EEG sensors given a hypothetical neural current distribution within the brain. This crucial step involves creating a realistic model of the head, a so-called volume conduction model (often just referred to as the head model), and a model of the potential source distribution, a so-called source model.
The source model defines the presumed location, orientation, and extent of neural activity. Common source models include distributed grids of current dipoles throughout the brain volume or dipoles constrained to the cortical surface. A head model describes the head's geometry and the electrical conductivity of its various tissues (e.g. scalp, skull, cerebrospinal fluid, brain). The complexity of the volume conductor model often differs between EEG and MEG. EEG typically requires more sophisticated, multi-layer models that accurately account for the distinct electrical conductivities of various head tissues, as these significantly influence the electric potential distribution measured on the scalp. For MEG, because magnetic fields are substantially less distorted by these different tissues, simpler volume conductor models (such as a single-shell model representing the brain or even a spherical approximation) can often be sufficient.
A common and effective approach is to construct these detailed head models using structural Magnetic Resonance Imaging (MRI) data, see Figure 7. This allows for the creation of accurate, individual-specific representations of head anatomy, significantly enhancing the precision of the forward model.

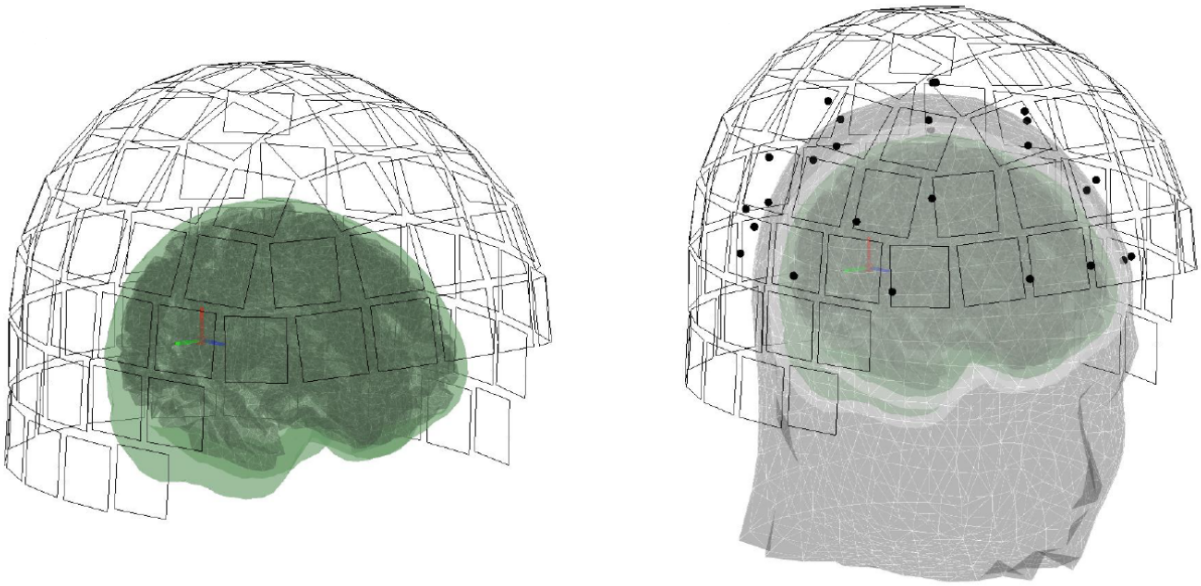


Figure 7: This figure illustrates the complete geometry for lead field calculation, derived from combined MEG and EEG measurements of a participant. It details the essential components required to solve the forward problem. Across both panels, a surface-based cortical source model is shown in black. A representative dipole position within this source model (located in the right transverse temporal gyrus) is highlighted, with axes indicating its potential orientations (x-axis: green, y-axis: blue, z-axis: red). The 102 MEG sensor triplets are depicted as square markers. **Left**: Shows the MEG-specific geometry, featuring a simple single-shell head model (representing the brain, shown in green). **Right**: Displays the combined geometry for simultaneous MEG and EEG. This setup shows the 32 EEG electrode positions (black dots) and utilizes a more elaborate three-compartment Boundary Element Method (BEM) head model, which is necessary for accurate EEG modeling. The BEM model delineates the scalp (dark gray), skull (light gray), and brain (green), with each compartment assigned a specific conductivity value.

A central concept in forward modeling is the lead field. Figure 8 illustrates an example for a dipole located in the right transverse temporal gyrus (see Figure 7). The lead field describes how a current source at a specific location and orientation within the brain contributes to the signal measured by each sensor. It represents the linear mapping between neural source activity and sensor measurements and is determined by the specific head model, source model, and sensor configuration used. In essence, the lead field matrix quantifies the sensitivity of each sensor to elemental current sources throughout the brain volume and is therefore crucial for predicting MEG/EEG signals from known neural activity.
Inverse modeling, in contrast to forward modeling, is the process of estimating the unknown neural sources that generated the recorded sensor data. This is typically the problem researchers address when analyzing

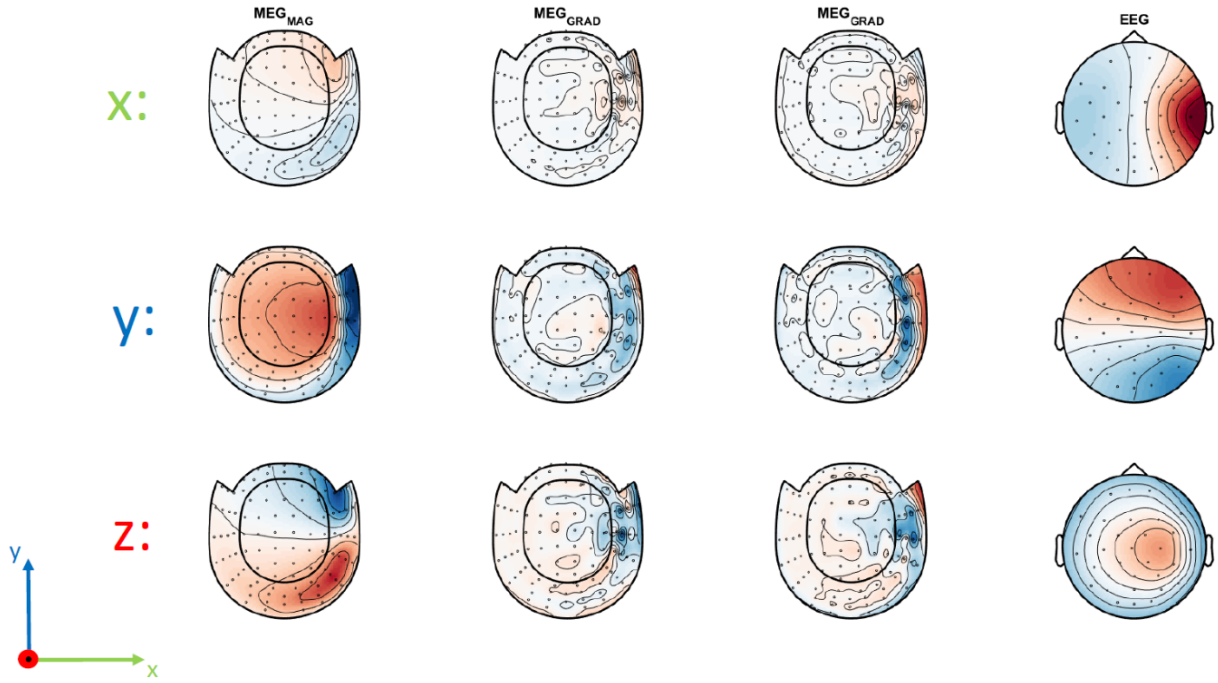


Figure 8: Lead field topoplots for a dipole located in the right transverse temporal gyrus. The dipole's location and color coding are consistent with those shown in Figure 7. These plots illustrate the magnetic field distribution (MEG) and electric potential distribution (EEG) corresponding to various dipole orientations. Each black dot marks a sensor location. Rows indicate the different dipole orientations, with the relevant right-handed coordinate system displayed at the bottom left (the z-axis points out of the plane). Columns specify the different sensor types; for clarity, MEG sensors are differentiated into magnetometers (MEGMAG) and two orthogonal planar gradiometer types (MEGGRAD). The lead fields for the MEG sensors were calculated using the single-shell head model, while those for the EEG electrodes were calculated using the BEM head model (see Figure 7).

experimental data. Inverse modeling is an ill-posed problem because infinitely many possible source configurations can produce the same sensor measurements. For instance, a radial source in magnetoencephalography (MEG) is inherently "invisible" to external sensors, meaning it can be added to any solution without altering the measured field. Therefore, additional constraints or assumptions must be introduced to find a meaningful or unique solution.

One widely used approach is dipole fitting, which assumes that a small number of Equivalent Current Dipoles (ECDs) can explain the measured data. The goal is then to find the optimal position, orientation, and strength of these ECDs such that the sensor data predicted by the forward model closely matches the observed data. This optimization process, commonly referred to as dipole fitting, relies heavily on the lead fields computed during forward modeling.

Together, forward and inverse modeling form the computational backbone of source localization in MEG and EEG. An accurate understanding of each component, particularly the lead field, is critical for the reliable interpretation of neuroimaging data. Chapter 3 will introduce the analysis pipeline for this project, detailing the computation of all the aforementioned concepts, including the source model, head model, and lead fields.

3 Methods: Experimental Procedure and Analysis

This section provides more detailed and focused theoretical background for the analysis presented in Chapter 4. The scripts required for this analysis are also provided within that chapter.

3.1 Experiment

3.1.1 Experimental Design

The experiment comprises three auditory stimulus conditions: clicks, up-chirps, and down-chirps. Each condition is presented 1200 times.

All stimuli are delivered at 40 dB HL (Hearing Level). To establish this level accurately for each participant, individual hearing thresholds for each of the three transient sound types are measured prior to the main experiment. A standard audiogram is also recorded to confirm normal hearing capabilities. Throughout the MEG recording sessions, participants watch silent movies of their choice to maintain a consistent level of alertness, but do not actively participate in the task.
The inter-stimulus interval (ISI) between consecutive tones is randomly varied between 350ms and 400ms. This timing results in approximately 8 minutes of recording per condition, leading to a total MEG data acquisition time of roughly 24 minutes for the stimuli, with the entire session lasting around 30 minutes including short breaks.
Additionally, a T1-weighted anatomical MRI scan is acquired for each participant to facilitate subsequent source localization analyses.

3.1.2 Participant Preparation for MEG Measurement

Prior to entering the magnetically shielded room (MSR), participants change into scrubs free of magnetizable parts and remove all magnetizable objects, such as jewelry.
To monitor physiological artifacts, Electrooculography (EOG) electrodes are placed near the eyes to record eye movements (blinks and saccades), and Electrocardiography (ECG) electrodes are attached to record cardiac activity. These concurrently recorded signals can be utilized during later data processing stages for artifact identification and removal.
A cap fitted with five Head Position Indicator (HPI) coils is placed on the participant's head. These coils enable continuous tracking of the participant's head position relative to the MEG sensors throughout the measurement. Accurate head position information is crucial because head movements within the MEG helmet can distort the recorded signals and introduce variance not attributable to the acoustic stimuli. Such movement-related effects must be minimized or compensated for.
If required by the specific study protocol, EEG electrodes are also prepared and applied at this stage.
Finally, for coregistration of the MEG data with the individual's anatomical MRI, the participant's head shape and the positions of key fiducial markers are digitized using a 3D digitizer pen (e.g., Polhemus Fastrak). Three anatomical landmarks—the nasion, and the left and right preauricular points—are digitized. These landmarks are essential for aligning the MEG coordinate system with the participant's MRI. Subsequently, the positions of the five HPI coils are digitized, followed by additional points distributed across the head surface to generate a comprehensive model of the participant's head shape. After the digitization process, the participant is carefully assisted into the MEG scanner.

3.2 Sensor-Level-Analysis: Computation of Auditory Event-Related Fields

3.2.1 Data Preprocessing and Epoching

To improve data quality, raw MEG recordings from Elekta systems are typically preprocessed using a tool called MaxFilter. This software enhances signal quality by removing environmental noise and correcting for head movements. A core feature of MaxFilter is the Signal Space Separation (SSS) algorithm, which isolates brain-originating signals from external magnetic interference. Additionally, MaxFilter allows to compensate for head motion by realigning the data to a consistent head position and identifies bad channels, which are then interpolated using neighboring sensors. (Taulu & Kajola, 2005; Taulu & Simola, 2006)
This part of the analysis focuses on identifying and visualizing the brain's responses to auditory stimuli, specifically MLAEFs. First, raw recordings of brain activity are preprocessed using filters to remove artifacts such as slow drifts, electrical interference, and high-frequency noise. Following this cleaning step, the continuous data is segmented into shorter time periods called epochs. Each epoch is created relative to (or "time-locked to") the onset of one of the short, transient sounds presented during the experiment. This process, known as epoching, is visualized in Figure 9.
To mark the timing of each sound within the recorded MEG signals, a trigger channel (STI101) is utilized. When a sound is presented, a short, synchronized pulse is sent via this channel and recorded alongside the MEG data. In practice, the sound and trigger are synchronized at the soundcard level. However, a slight latency ($<$ 10ms) can exist between the recorded MEG signals and the trigger. This discrepancy arises because the sound travels through insert earphone tubes, taking a fraction of time to reach the ear, and there's also a processing delay for MEG signals compared to the directly recorded trigger. To ensure the timing of brain responses accurately reflects when the sound was perceived, a correction can be applied to compensate for this sound-trigger delay.
After epoching, the data segments are inspected for artifacts, such as eye blinks, heartbeats, general muscle activity, or other unexpected events. Highly contaminated epochs can be rejected, either through visual in-

spection or by automatic flagging based on predefined thresholds. This results in a dataset of clean epochs for further analysis.

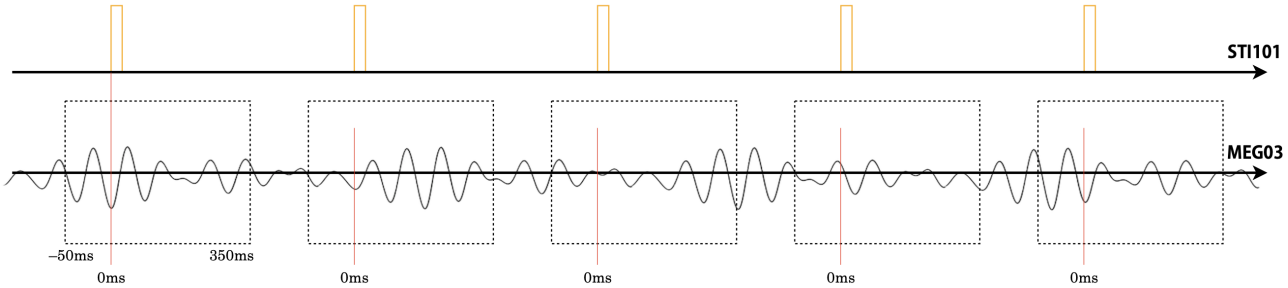


Figure 9: Illustration of MEG data epoching relative to stimulus trigger onsets. Displayed are an example MEG channel (MEG03) and the stimulus trigger channel (STI101), which marks stimulus presentation times with short retangular pulses.

3.2.2 Averaging and Event-Related Responses

Next, the brain responses to repeated presentations of the same sound are averaged. This averaging process enhances consistent, time-locked neural patterns while minimizing random noise, thereby increasing the signal-to-noise ratio. The underlying assumption is that the brain produces a consistent, time-locked response to the sound, which becomes more prominent as multiple epochs are averaged, while signals unrelated to the acoustic events tend to average out.
This averaging process yields what are known as event-related responses (or evoked fields for MEG). Following this pipeline, an event-related response will be available for each of the 306 MEG channels. While inspecting all individual channels can be laborious, these sensor-level results provide the initial foundation for more detailed investigations, such as determining the specific brain regions involved in processing auditory information.

3.3 Source-Level-Analysis

3.3.1 Forward Modeling

This section details the construction of the forward model for this analysis, referencing the general principles outlined in Chapter 2.5.2. The primary objective is to compute the lead fields. Lead fields mathematically describe how a specific source (equivalent current dipole) in a volume conductor is spatially projected onto the MEG/EEG sensors, e.g. the spatial potential or field distribution for a given source. Accurate lead fields are essential for the subsequent step of solving the inverse problem: projecting measured sensor data back to the brain to estimate neural source activity. The creation of the forward model involves developing both a head model (volume conductor model) and a source model, both of which are typically derived from each participant's anatomical MRI.

MRI Processing and Co-registration: The process begins with the participant's structural T1-weighted MRI scan, which provides a detailed anatomical image. To align the MRI with the MEG sensor geometry, a coregistration procedure is performed. This step utilizes the three digitized anatomical landmarks (nasion, left and right preauricular points) and the set of additionally digitized head shape points recorded prior to the MEG session. Successful coregistration allows the participant's anatomy and the MEG sensor data to be expressed in a common head coordinate system, defined by these anatomical landmarks. After alignment, the MRI scan is typically resliced into a uniform 3D grid to ensure compatibility with subsequent analysis steps.

Head Model (Volume Conduction Model): Following coregistration, the MRI data is used to create the volume conduction model. This involves segmenting the MRI into different tissue types (e.g., scalp, skull, brain), each characterized by distinct electrical conductivity values. For MEG analysis, a simpler head model comprising a single compartment representing the brain may suffice, as magnetic fields are relatively unaffected by the skull and scalp. In contrast, EEG analysis typically requires a more detailed multi-compartment model (e.g., at least scalp, skull, and brain) because electrical potentials are significantly distorted by these tissues, especially smeared by the skull. The head model mathematically describes how electromagnetic fields propagate from the neural sources through these various head tissues to the sensors.

Source Model: Finally, the source model, which defines the set of possible locations and orientations of neural activity, is created, often based on the segmented brain tissue (an example is shown in Figure 10). Two primary approaches for defining the source model are common:

1. **Volume-based source model:** This approach defines potential source locations as a grid of dipoles distributed throughout the brain volume. Typically, these grid positions are initially defined in a standardized brain space (e.g., Montreal Neurological Institute (MNI) – coordinates) and are then non-linearly warped to fit each participant's specific anatomy. This ensures that the grid points correspond to relatively consistent, anatomically-defined locations across subjects.
2. **Surface-based source model:** Alternatively, the source locations can be constrained to the cortical surface. This involves tessellating the cortical surface (usually the grey-white matter boundary extracted from the MRI) into a mesh of vertices. Dipoles are then placed at these vertices, often with their orientation constrained to be perpendicular to the cortical surface.

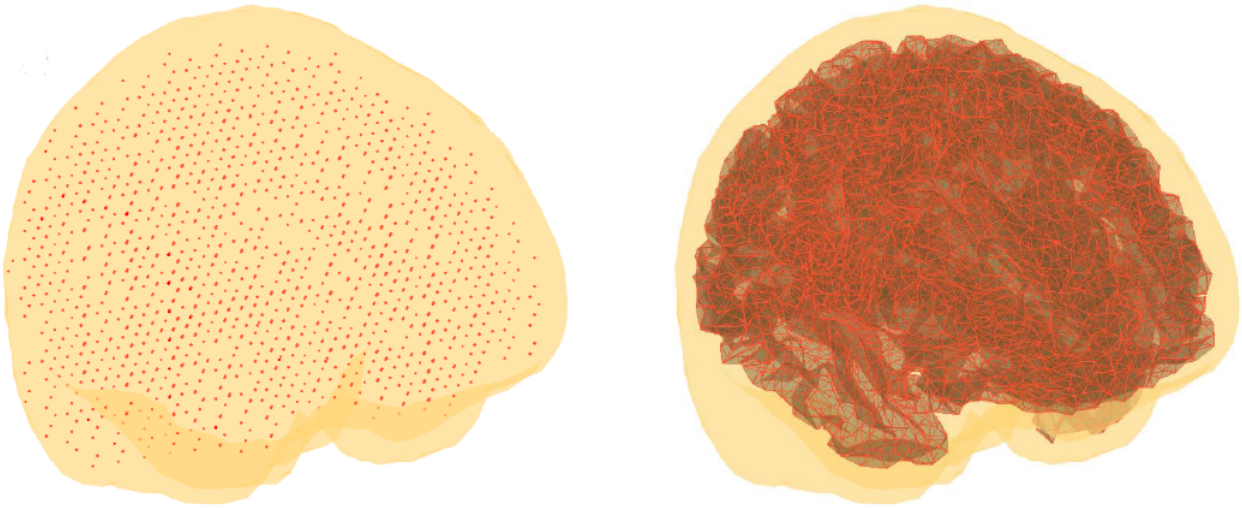


Figure 10: Head and source models for a single participant. The figure displays a single-shell head model (yellow) alongside two distinct source models (red). **Left:** A volume-based source model structured as a regular 3D grid within MNI coordinates. This model contains 1459 dipole positions. **Right:** A surface-based source model derived from a tessellation of the cortical surface generated using FreeSurfer and HCP Workbench. This model features 4000 dipole positions per hemisphere.

Throughout these model construction steps, visual checks are performed to verify the successful alignment and segmentation of the data. An accurately constructed forward model is an essential prerequisite for subsequent source localization, as it enables the estimation of neural activity within the brain from the signals recorded by external sensors.

3.4 Dipole Modeling and Source Waveform Estimation

This analysis step focuses on identifying and characterizing the estimated locations and properties of the underlying brain activity sources, modeled as Equivalent Current Dipoles (ECDs). These ECDs are mathematical representations that help describe the likely origin and temporal behavior of the brain's electrical currents generating the measured MEG signals (see Section 2.3).

3.4.1 Initial Two-Dipole Symmetric Model and Refinement

Because auditory stimuli were presented to both ears, a symmetrical two-dipole model is assumed as an initial step. Symmetry, in this context, implies that the dipoles are constrained (e.g. to have the same magnitude for the x-component of their dipole location, reflecting the brain's bilateral symmetry; see Figure 8 for coordinate axes). The initial dipole fit is based on the N19m–P30m component of the MLAEFs.
This symmetry-constrained two-dipole model is fitted first. Subsequently, the fit is refined by relaxing the symmetry constraint and locally optimizing the dipole locations, orientations, and strengths.
For the initial fit, data from the down-chirp and click conditions are pooled. This doubles the number of epochs used for the MLAEF computation, thereby increasing the signal-to-noise ratio. The dipole locations and orientations identified from this pooled data are then used as a basis for a spatial filter to estimate the dipole time

courses for each of the three conditions (click, up-chirp, and down-chirp) separately. The positions are held constant over conditions and should ideally be localized in the PAC.

3.4.2 Spatial Filtering Options

There are different ways to use the fitted two-dipole model as a spatial filter to extract source waveforms:

1. **Fixed Location, Free Orientation:** The dipole locations are fixed, but their orientations are allowed to vary freely over time. Therefore, the x, y, and z components of the dipole moments are estimated for all conditions, yielding three time courses per dipole.
2. **Fixed Location, Fixed Orientation (SVD-based):** The dipole positions and orientations are fixed over time for each condition. The fixed orientation is determined by applying a singular value decomposition (SVD) to the initially estimated x, y, and z dipole moment components within the N19m–P30m time window. The orientation explaining the maximum variance is selected and held constant. This approach returns a single time course representing the amplitude along this SVD-based orientation for each dipole.
3. **Fixed Location, Fixed Orientation (Mean-based):** The dipole positions and orientations are fixed over time for each condition. The fixed orientation is based on the previously computed dipole moment components (x, y, z), and the mean orientation is calculated over the N19m–P30m time window. This method returns a single time course representing the amplitude along this mean orientation for each dipole.

It is recommended to use **Option 3 (fixed dipole with mean orientation)** for further analysis because it is simple, yielding one time course per dipole (i.e., two time courses for the dipole pair).

3.4.3 Visualization and Anatomical Localization

Initially, estimated dipole locations and orientations are visualized on each participant's anatomical MRI to verify their plausibility and bilateral symmetry. An example of this visualization is provided in Figure 11.

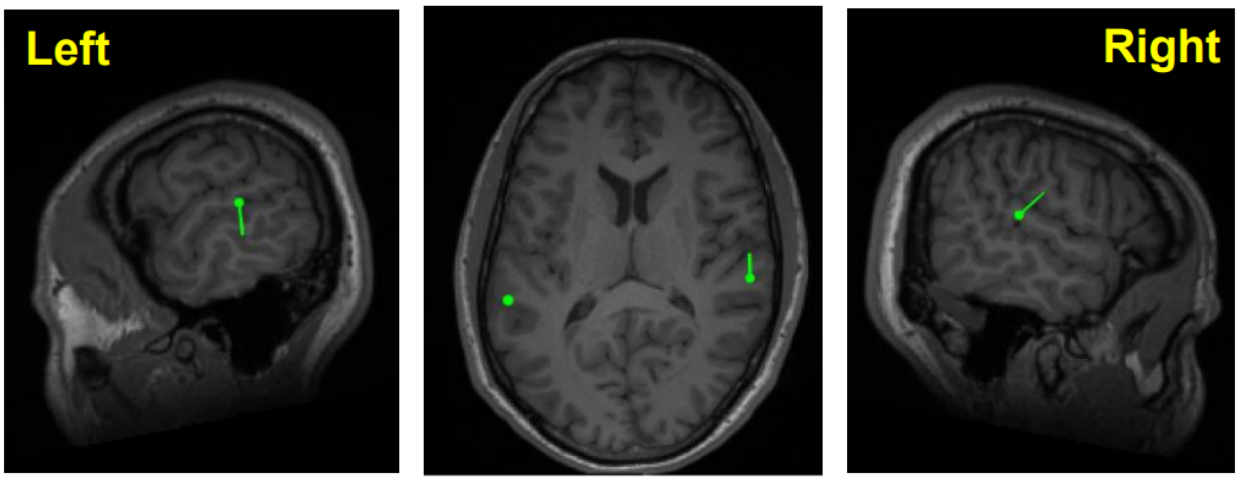


Figure 11: Dipole source locations in the left and right temporal lobes. Example from a single participant.

Next, the dipole locations are transformed into the common MNI coordinate space. This transformation allows for consistent comparisons across participants and facilitates anatomical labeling. Each dipole location is then matched with the nearest brain region using a standard anatomical atlas, aiding in the interpretation of which brain areas are involved in processing the auditory stimuli.
The dipole moment time courses (also known as source waveforms) are then plotted, illustrating how the estimated source activity evolves over time in response to the stimuli. The time courses for the three stimulus conditions can subsequently be compared. An exemplary time course from one hemisphere is depicted on the cover.

3.4.4 Summary

Overall, this dipole modeling step allows for a detailed inspection of the estimated brain sources: how well they are localized, their anatomical plausibility, and how their activity changes over time. This information supports a more reliable interpretation of the neural mechanisms underlying the signals recorded at the sensor level.

3.5 Data Organization and Naming Convention

The organization and naming of all recordings adhere to the Brain Imaging Data Structure (BIDS). To read the full specification, find tools, and view example datasets, refer to the official BIDS website.

3.5.1 Directory Structure

All raw data is stored within the `rawdata/` directory. Each participant has a dedicated folder, identified by a unique subject number (`XY`), and the internal structure is as follows:

- `rawdata/`
  - `sub-01/`
    - `anat/` (Anatomical MRI data)
    - `beh/` (Behavioral data)
    - `meg/` (MEG recordings)
  - `sub-02/`
    - `...`
  - `...`

3.5.2 File Naming Conventions

All filenames must strictly follow the patterns outlined below.

- **Anatomical Data (anat)**: The T1-weighted MRI scan must be named:

  `sub-XY_T1w.nii`

  Note: Original DICOM files must be converted to the NIfTI format (`.nii`). This can be accomplished using tools such as MRIcron's dcm2niix. For software and documentation related to DICOM conversion, visit the official MRIcron page on the Neuroimaging Tools & Resources Collaboratory (NITRC).

- **Behavioral Data (beh)**: This directory contains supplementary data, such as the participant's hearing thresholds.

- **MEG Data (meg)**:
  - Task recordings are named according to the pattern:

    `sub-XY_task-taskname_meg.fif`

    where `taskname` corresponds to the task performed and must be one of the following: `click`, `up`, or `down`.
  - The empty room recording for each subject must be named:

    `sub-XY_task-emptyroom_meg.fif`

4 Data Analysis using FieldTrip

In this study, you will work with preprocessed MEG data that has undergone MaxFilter. Your analysis will build upon this data. You will be responsible for all subsequent steps, including computing Middle-Latency Auditory Evoked Fields (MLAEFs), creating the head and source models, performing dipole fitting, and generating all associated visualizations.
The analysis code for the data, along with the experimental paradigms and additional documentation, can be downloaded from the following GitHub repository: https://github.com/tillhabersetzer/meg-practical.git.

4.1 Analysis Pipeline

The analysis pipeline must be configured correctly. Therefore, you first need to download and set up the latest version of the FieldTrip toolbox (Oostenveld et al., 2011). Installation instructions can be found on the FieldTrip website. Alternatively, please consult the instructor for assistance in correctly setting up FieldTrip.
For the analysis pipeline provided to you, your primary configuration task is to adjust the `settings_analysis.m` file. Within this file, you must define the paths to your FieldTrip toolbox installation and the main project directory:

- `settings.path2fieldtrip`
- `settings.path2project`

You will be provided with a project folder containing scripts for all necessary analysis steps. Your task is not to implement these analysis scripts from scratch, as this would be excessively complex for the scope of this practical part. Instead, your responsibility is to execute the provided scripts, understand the core concepts underpinning each analysis step, and interpret the resulting data.

4.1.1 Sensor-level Analysis

The primary scripts for this stage of analysis are listed in Table 2. Each script includes comments explaining its purpose and operations.

Script Name	Description
`compute_erfs.m`	Computes Middle-Latency Auditory Evoked Fields (MLAEFs) for each participant.
`plot_erfs.m`	Visualizes the MLAEFs computed by `compute_erfs.m`.

Table 2: Overview of scripts for sensor-level MLAEF computation and visualization.

The main computational work for generating MLAEFs for each subject is performed by `compute_erfs.m`. This script executes the following operations for each participant (see Chapter 3.2 for more theoretical details):

1. Filters the continuous MEG data using a band-pass filter and a notch filter.
2. Defines epochs around auditory stimuli (clicks and chirps) based on trigger events. Additionaly, a pooled condition combining cicks and down-chirps is computed for the dipolfit.
3. Applies an optional latency correction to account for discrepancies between audio playback timing and trigger signal onset.
4. Segments the data into epochs and optionally performs semi-automatic artifact rejection through visual inspection (an example of epoching is shown in Figure 9). Alternatively, artifact rejection can be automated based on predefined thresholds.
5. Computes time-locked averages (MLAEFs) for each stimulus condition.
6. Saves the resulting averaged MLAEFs and associated trial information for subsequent visualization and source-level analysis.

The output of `compute_erfs.m` consists of condition-specific, sensor-level MLAEFs, which form the basis for subsequent analyses. Depending on the number of subjects and chosen preprocessing options, script execution may take several minutes to complete. The results can then be visualized by executing `plot_erfs.m`. This visualization script allows for the inspection of topographic maps and time series of the MLAEFs for each participant, as well as data from individual sensors. It facilitates the comparison of peak amplitudes and latencies of the MLAEFs across conditions.

4.1.2 Source-level Analysis

Forward Modeling In this step, the head model and source model, which are prerequisites for the subsequent dipole fit, are computed. These models are derived from each participant's recorded anatomical T1-weighted MRI scan. Refer to Chapter 2.5.2 and Figure 10 for more theoretical details and examples.
The `headmodel.m` script performs the following key steps for each participant:

1. Extracts the digitized headshape information from the MEG dataset.

Script Name	Description
`headmodel.m`	Handles MRI import, coregistration with MEG sensor geometry, MRI reslicing and segmentation, head model construction (volume conduction model), saving outputs, and includes visual checks for quality control.
`volumetric_sourcemodel.m`	Computes a volume-based source model. This involves defining source locations as a grid of dipoles in a standardized brain space (e.g., MNI) and non-linearly warping this grid to fit each participant's specific anatomy. Outputs are saved, and visual checks are included.

Table 3: Overview of scripts for head model and source model computation.

2. Reads the participant's T1-weighted anatomical MRI.
3. Co-registers the MRI with the MEG (e.g., Neuromag) coordinate system. This process typically uses digitized anatomical landmarks (nasion, preauricular points) for initial alignment, followed by refinement using the digitized headshape data via an Iterative Closest Point (ICP) algorithm.
4. Reslices the coregistered MRI to a uniform voxel grid to ensure compatibility with subsequent processing steps.
5. Segments the resliced MRI to extract relevant tissue compartments (e.g., brain volume) required for forward modeling.
6. Constructs a volume conduction model (head model), often using the single-shell approach (which is generally sufficient for MEG analysis).
7. Saves all intermediate and final outputs (e.g., coregistered MRI, resliced MRI, segmented MRI, and the head model) for quality control and subsequent use in source reconstruction.

Throughout these steps, visual checks are incorporated to ensure the quality and accuracy of alignment and segmentation. The successful creation of these models is an essential prerequisite for computing the lead field and subsequently performing source reconstruction (i.e., estimating neural activity in source space from sensor-level data).

Dipole Fitting and Source Waveform Analysis In this step, using the previously computed MLAEFs, source model, and head model, a dipole fitting procedure is performed on the data to estimate neural source activity. The main computational work is carried out by `dipolefitting.m`. This script executes a series of operations for each participant (see Chapter 3.3.1 for more theoretical details). The resulting source activity can then be visualized by executing `plot_dipolfitting_subjectlevel.m`. An overview of these scripts is provided in Table 4.

Script Name	Description
`dipolefitting.m`	Performs an initial symmetry-constrained dipole fit on pooled condition data, which is subsequently refined without the symmetry constraint. Based on the estimated dipole locations, it then computes and saves dipole moment time courses for each individual stimulus condition using various spatial filtering approaches.
`plot_dipolfitting_subjectlevel.m`	Visualizes estimated dipole locations (in individual MRI and standard MNI space), maps these locations to anatomical labels using an atlas, and plots the dipole moment time courses derived from different spatial filtering options and for all stimulus conditions (further details on options in Chapter 3.4.2).

Table 4: Scripts for dipole fitting and source-level analysis.

The script `dipolefitting.m` performs the following key tasks for each participant:

1. Loads the precomputed head model, source model, and MLAEFs.
2. Based on the N19m–P30m component of the MLAEFs, performs an initial symmetry-constrained two-dipole fit (one dipole per hemisphere) using data pooled from the down-chirp and click conditions.

- The fit can be performed using either magnetometer or gradiometer data; gradiometers are recommended for this study. Combination of sensor types, though possible, is not utilized here.
- This initial dipole fit is then locally refined by relaxing the symmetry constraint.

3. Estimates the dipole moment time courses for the individual click, up-chirp, and down-chirp conditions using the refined dipole locations as a spatial filter. Several options for this estimation are implemented:
   (a) *Fixed Position, Free Orientation:* Dipole locations are fixed, while orientations are allowed to vary freely over time (estimating x, y, z dipole moment components).
   (b) *Fixed Position, Fixed Orientation (SVD-based):* Dipole locations are fixed, and orientation is fixed to the direction explaining maximum variance (derived via Singular Value Decomposition of the initial x, y, z moments within the N19m–P30m window). This yields a single amplitude time course per dipole.
   (c) *Fixed Position, Fixed Orientation (Mean-based):* Dipole locations are fixed, and orientation is fixed to the mean orientation computed from the initial x, y, z moments over a specific time window (e.g., N19m–P30m). This also yields a single amplitude time course per dipole.

   (Refer to Chapter 3.4.2 for more details on these spatial filtering options.)
4. Saves the estimated dipole positions and the various computed time courses for subsequent visualization and analysis.

The `plot_dipolfitting_subjectlevel.m` script facilitates detailed exploration of these results by performing the following visualization tasks:

1. *Dipole Localization Visualization:* Displays the fitted dipole positions and orientations within each subject's anatomical MRI space for both the initial symmetry-constrained and the refined (non-symmetric) dipole fits. This allows for visual assessment of source placement accuracy and hemispheric symmetry.
2. *Anatomical Mapping and Labeling:* Transforms the dipole positions into standard MNI space and identifies the nearest anatomical regions using a predefined atlas (e.g., AAL atlas), aiding in the neuroanatomical interpretation of the source locations and facilitating comparability across subjects.
3. *Source Waveform Analysis:* Plots the dipole moment time courses for each stimulus condition and hemisphere. This includes visualizations of:
   - The x, y, and z components from freely oriented dipole models.
   - The single amplitude time courses from fixed-orientation models (SVD-based and mean-based).

   These plots provide insights into the temporal evolution and directional characteristics of the estimated source activity.
4. *Comparison Across Conditions:* Facilitates the comparison of peak amplitudes and latencies of the source waveforms across the different stimulus conditions. It is recommended to use the fixed-orientation time courses (e.g., based on mean orientation) for simpler visual interpretation in such comparisons.

Overall, these analysis and visualization steps enable a thorough inspection of the dipole fitting quality, its anatomical plausibility, and the resulting source dynamics, supporting a reliable interpretation of the MEG source localization results.

5 Practical Report

General Report Structure:

- Clear, concise report: Introduction, Theory, Methods, Results, Discussion.
- Proper labeling, captions, citations and referencing for all figures/tables.

Experiment Overview & Theory:

- **Objective & Hypotheses:**
  - State experiment goal, name hypothesis
    1. Higher peripheral synchronization (chirps) yields larger N19m-P30m complex.
    2. N19m-P30m latency depends on stimulus type (clicks vs. chirps).
- **Key Theory:**
  - **Anatomy:** Relevant auditory pathway to cortex, N19m-P30m generation.
  - **Experiment:** Stimulus design and link to anatomy. Clicks vs. chirps, explain peripheral synchronization compensation by chirps.
  - **MEG:** Principles of magnetic field measurement from neural activity.
  - **Design:** Within-subject comparison of stimulus types.

Experimental Setup & Data Acquisition:

- **Preparation:** Subject preparation (ECG/EOG, landmarks, HPI coils, head shape).
- **Additional measurements:** Audiogram, thresholds for clicks/chirps, Anatomical MRI and link to recordings.
- **MEG Setup:** Positioning in MEG, MEG setup (earphone attachment, video etc.).
- **MEG Data:** Record data for all 3 conditions (up-chirp, down-chirp, click).

Data Analysis:

- **Describe all analysis steps:** filtering, artifact removal, epoching, baseline correction, averaging, dipole modeling.
- **Execute analysis pipeline:** Compute MLAEFs, head model (from MRI), sourcemodel, and dipole fit, report problems.
- **Quality Control:** Check data quality, dipole location/timecourse.

Results & Discussion:

- **Comparisons:** Compare condition time courses in sensor and source space. Note differences.
- **Visualization:** Plot informative results (sensors, waveforms, topographies, source activity).
- **Hypothesis Testinga and Interpretation:** Discuss findings in relation to the hypotheses. Interpret results, discuss limitations.

6 References

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